

**FMUSER FUQ-4665 HD-SD Encoder Modulator 4 Input Channels
HDMI-YPbPr-CVBS to DVB-T-C-ATSC RF MPEG-2 MPEG-4 AVC
H.264 Encoding USB Record Playback Broadcast Headend**



HDMI/CVBS/YPbPr  **DVB-T/DVB-C/ATSC RF**

--- Home Version



A device that allows audio and video signal input in TV distributions and video recorded and playback through a USB port, with applications in home entertainment, Digital Signage for hotels, shops, restaurants etc...

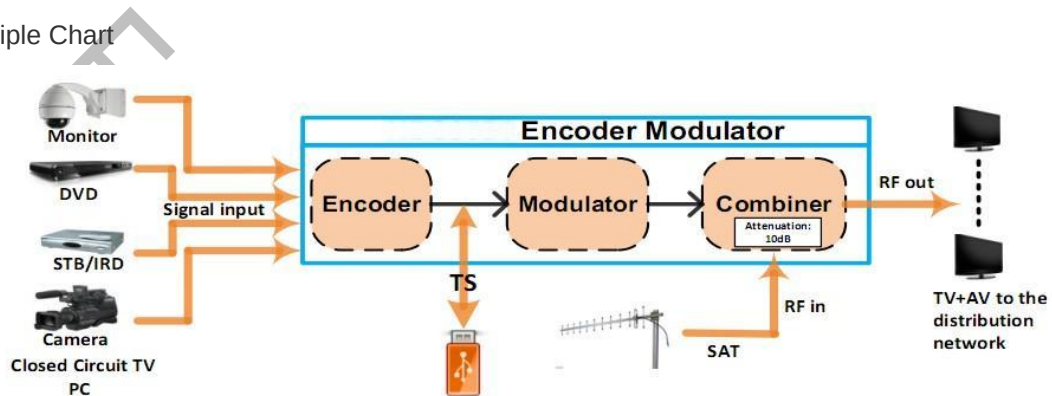
Product Overview

The FUQ-4665 performs MPEG-2/MPEG-4 AVC encoding and DVB-T/C/ATSC modulation with HDMI/YPbPr/CVBS inputs. It processes HD/SD signals into digital RF output (30-960MHz) with USB record/playback capability. This 1RU rack-mount unit supports NMS control via Ethernet. Ideal for digital terrestrial broadcasting and multi-screen headend systems requiring low-latency conversion.

Key feature

Dolby Digital AC3 2.0 audio pass-through via HDMI input

Principle Chart



Technical Specifications

Encoding Section	Encoding Section	Encoding Section	Encoding Section
HDMI	Video	Encoding	MPEG-2, MPEG-4 AVC/H.264
HDMI	Video	Interface	HDMI*1
HDMI	Video	Resolution	1920*1080_60P, 1920*1080_50P à (for MPEG-4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i; 1280*720_60p, 1280*720_50P
HDMI	Video	Bit rate	0.500~19.500 Mbps
HDMI	Audio	Encoding	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC; DD AC3 2.0 (Optional) AC3 Passthrough
HDMI	Audio	Interface	HDMI
HDMI	Audio	Sample rate	48KHz

HDMI	Audio	Bit rate	64, 96, 128, 192, 256, 320, 384 kbps
YPbPr/ CVBS	Video	Encoding	MPEG-2, MPEG-4 AVC/H.264
YPbPr/ CVBS	Video	Interface	CVBS *1, YPbPr*1
YPbPr/ CVBS	Video	Resolution	CVBS: 720x576_50i (PAL); 720x480_60i (NTSC) YPbPr: 1920*1080_60i, 1920*1080_50i; 1280*720_60p, 1280*720_50P
YPbPr/ CVBS	Video	Bit rate	0.500~19.500 Mbps
YPbPr/ CVBS	Audio	Encoding	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC; DD AC3 2.0 (Optional)
YPbPr/ CVBS	Audio	Interface	1 Stereo /2 mono
YPbPr/ CVBS	Audio	Sample rate	48KHz
YPbPr/ CVBS	Audio	Bit rate	64, 96, 128, 192, 256, 320, 384 kbps

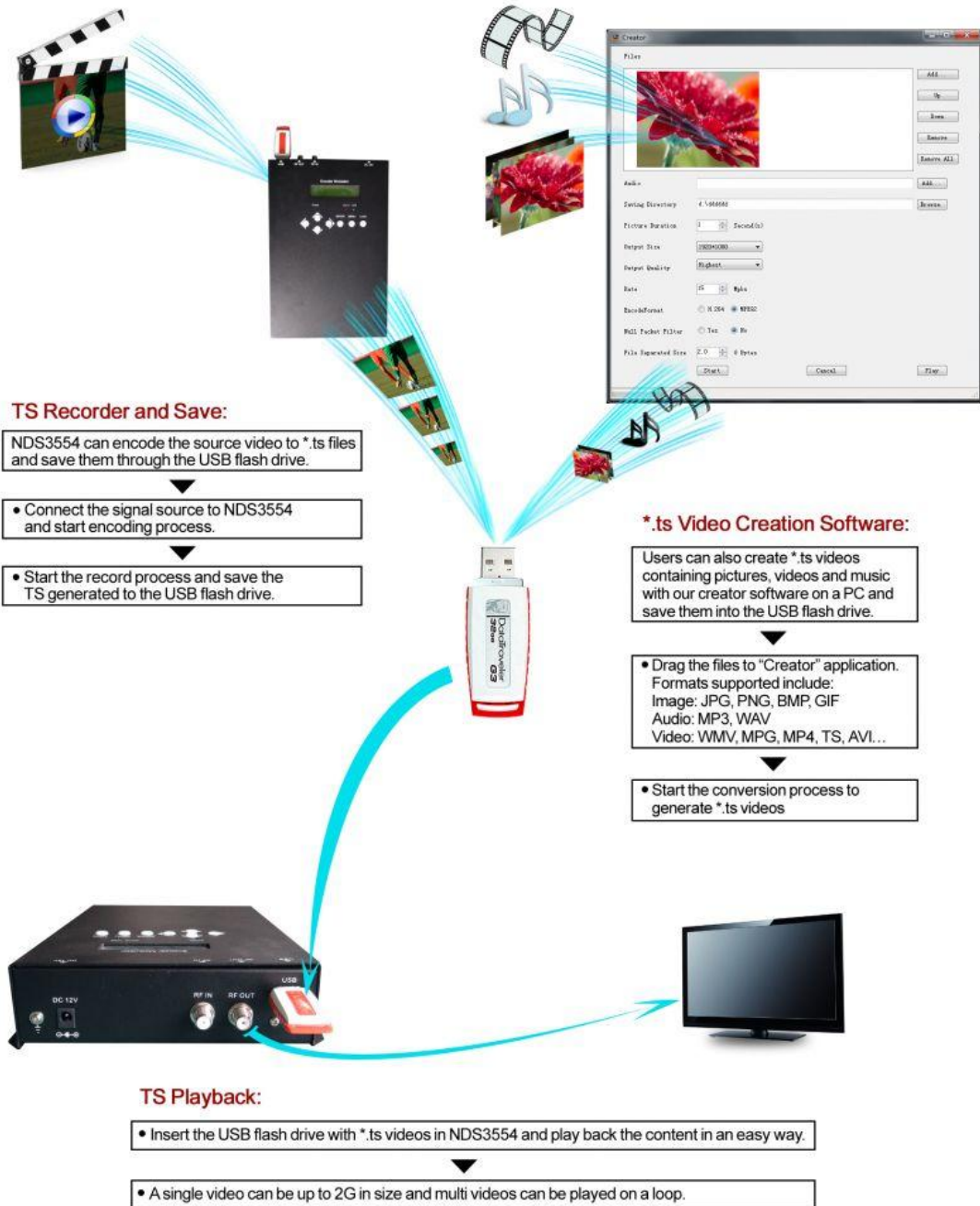
DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section	DVB-T Modulator Section
Standard	Standard	Standard	DVB-T COFDM	DVB-T COFDM	DVB-T COFDM	DVB-T COFDM
Bandwidth	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM,	QPSK, 16QAM, 64QAM,	QPSK, 16QAM, 64QAM,	QPSK, 16QAM, 64QAM,
Code rate	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.	1/2, 2/3, 3/4, 5/6, 7/8.
Guard Interval	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Transmission Mode:	Transmission Mode:	Transmission Mode:	2K, 8K	2K, 8K	2K, 8K	2K, 8K

MER	MER	MER	≥42dB	≥42dB	≥42dB	≥42dB
RF frequency	RF frequency	RF frequency	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step
RF output level	RF output level	RF output level	-16~ -36 dBm (71~91 dBμV), 0.1db step	-16~ -36 dBm (71~91 dBμV), 0.1db step	-16~ -36 dBm (71~91 dBμV), 0.1db step	-16~ -36 dBm (71~91 dBμV), 0.1db step
DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)	DVB-C Modulator Section (Option)
Standard	J.83A (DVB-C)	J.83A (DVB-C)	J.83A (DVB-C)	J.83A (DVB-C)	J.83B	J.83C
Constellation	16/32/64/128 /256 QAM	16/32/64/128 /256 QAM	16/32/64/128 /256 QAM	16/32/64/128 /256 QAM	64/ 256 QAM	64/ 256 QAM
Bandwidth	8M	8M	8M	8M	6M	6M
MER	≥43dB	≥43dB	≥43dB	≥43dB	≥43dB	≥43dB
RF frequency	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step
RF level	-16~ -36dbm(71~91 dBμV), 0.1db step	-16~ -36dbm(71~91 dBμV), 0.1db step	-16~ -36dbm(71~91 dBμV), 0.1db step	-16~ -36dbm(71~91 dBμV), 0.1db step	-16~ -36dbm(71~91 dBμV), 0.1db step	-16~ -36dbm(71~91 dBμV), 0.1db step
Symbol rate	5000-9000 Ksps	5000-9000 Ksps	5000-9000 Ksps	5000-9000 Ksps	5000-9000 Ksps	5000-9000 Ksps
ATSC Modulator Section (Option)	ATSC Modulator Section (Option)	ATSC Modulator Section (Option)	ATSC Modulator Section (Option)	ATSC Modulator Section (Option)	ATSC Modulator Section (Option)	ATSC Modulator Section (Option)
Standard	Standard	ATSC A/53	ATSC A/53	ATSC A/53	ATSC A/53	ATSC A/53
MER	MER	≥42dB	≥42dB	≥42dB	≥42dB	≥42dB
RF frequency	RF frequency	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step	30~960MHz, 1KHz step

RF output level	RF output level	-16~ -36 dBm (71~91 dbμV), 0.1db step	-16~ -36 dBm (71~91 dbμV), 0.1db step	-16~ -36 dBm (71~91 dbμV), 0.1db step	-16~ -36 dBm (71~91 dbμV), 0.1db step	-16~ -36 dBm (71~91 dbμV), 0.1db step
Constellation	Constellation	8VSB	8VSB	8VSB	8VSB	8VSB
System	System	System	System	System	System	System
Management	Management	Management	Management	Local control: LCD + control buttons	Local control: LCD + control buttons	Local control: LCD + control buttons
Language	Language	Language	Language	English	English	English
LCN Insertion	LCN Insertion	LCN Insertion	LCN Insertion	yes	yes	yes
General	General	General	General	General	General	General
Power supply	Power supply	Power supply	Power supply	DC 12V	DC 12V	DC 12V
Dimensions	Dimensions	Dimensions	Dimensions	144*238*52 mm	144*238*52 mm	144*238*52 mm
Weight	Weight	Weight	Weight	Approx 1kg	Approx 1kg	Approx 1kg
Operation temperature	Operation temperature	Operation temperature	Operation temperature	0~45■	0~45■	0~45■



USB Recoder & Player



USB Flash Drive Specifications Required:

- Standard: High Speed 2.0
- File System: FAT 32
- Memory: 32G is suggested

Typical Applications

...for communities of residents
an information channel on their television



...for restaurants
information about daily menus ,special deals,etc



...for hotels
meeting rooms,exhibitions,message,etc



...for hospitals
training courses, healthy guide, etc



...for Public Spaces
advertising,user information,news,etc



...for shopping centres
new collections,special deals,etc



**Create your own advertising and information
channel using only your NDS3554 Encoder Modulator**

Technical Compare

Main Features	FMUSER FU-4665/FU-4667	Competing Products
Encode	I MPEG2 HD/SD; MPEG4 HD/SD	I MPEG2 SD
Record TS to USB memory	I Yes (FU-4665&FU-4667)	I No
*.ts files play-mode	I Both Multi/Single files loop I Support HD&SD video play	I Single file loop I Support only SD video play

TS Creator.exe	I HD video to HD video transform I Null Packet Filter, effect bitrate up to 90% I Support output video size, rate, and format control	I HD video to SD video downscale I No Null Packet Filter, effect bit rate only around 50%
Single video length	I 2-3 Hours HD/SD	I 40 Minutes SD
Low Delay	I Supported	I Not supported
DD AC3 (2.0) audio encode	I Supported	I Not supported

Internal Test Report of Latency

The values cover the progress from Encoding to Decoding

Decoding Terminal	Encoding Details					Average Latency (ms)
	Single Source Interface	Bit Rate Mode	Resolution	DTS Mode	Encoding Type	
DVB-T HD STB	HDMI	VBR	1080i@50	DTS=1 (Mode 1)	mpeg2	300
					H.264	335
				DTS=40 (Mode 2)	mpeg2	407.5
					H.264	492.5
			720p@50	DTS=1 (Mode 1)	mpeg2	230
					H.264	285
				DTS=40 (Mode 2)	mpeg2	382.5
					H.264	395
DVB-T HD STB	YPbPr	VBR	1080i@50	DTS=1 (Mode 1)	mpeg2	282.5
					H.264	395
				DTS=40 (Mode 2)	mpeg2	397.5
					H.264	450
			720p@50	DTS=1 (Mode 1)	mpeg2	267.5
					H.264	255
				DTS=40 (Mode 2)	mpeg2	385
					H.264	422.5
DVB-T HD STB	CVBS	VBR	576i@50	DTS=1 (Mode 1)	mpeg2	450
					H.264	570
				DTS=40 (Mode 2)	mpeg2	510
					H.264	620

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)